

HAWKEYE PIPEPRO INSTALLATION PROCEDURE

The instructions that follow explain the procedure for installing the PipePro III Machine and its related components. Some final assembly procedures are to be performed only under the direct supervision of the Hawkeye technician. Those procedures requiring Hawkeye's supervision are noted in these instructions. Follow the procedures in the order that they are listed. Refer to the illustrated aids as instructed. Please note that most procedures call out for tack welding after assembly. The purpose of this is to allow our technician to inspect the installation before final welding. In the unlikely event that something needs corrected, the modification can be made easier if the welds are only tacked.

INSTALL PIT STILT POST

1. Clean pit of all foreign materials and recheck foundation to blueprint to be assured of proper constructions.
2. Clean excess concrete off of and around all bearing pads. Pads must be clean in order to apply an effective weld. Now is the best time to clean the pads as it is impossible to clean the pads after the machine and stilt post are in place.
3. If not already done, weld the (2) 1" plates (E) to the (4) 5" iron pipe steel legs (B) as specified on the foundation blueprints.
4. Position the Hawkeye-supplied 6" channel/6" post assembly into the pit at the weld pads near the pressurehead station. A height-adjusting collar is required at the base of each 6" pipe leg. Do not attempt to final weld or position the height of the support assembly at this time. It will be set after the machine has been positioned in the pit.
5. Using a transit, set the level at each of the stilt leg tops except for the (2) center stilt legs (A) which will be installed in a later step, to within $\pm 1/16"$ and all are plumbed vertical. Each stilt leg is to have a height-adjusting collar at the base of each leg. Refer to the foundation blueprints for the proper stilt installation procedure.
6. When the stilt legs are located as specified by the foundation blueprints, and top of legs to within $\pm 1/16"$ and plumbed to vertical, then weld to the anchor pads as specified by the foundation blueprints.
7. Mark the center point of the pit on the pit floor. This point will be referred to in later procedures when positioning the machine base.

INSTALL MACHINE BASE/TURNTABLE

1. Following the handling instructions for lifting the machine base/turntable assembly, set the assembly into the pit allowing the base to bear upon the stilt post tops.

2. With the base bearing upon the stilt posts, and still secured to the lifting means, send a man into the pit under the base to check initial alignment of the machine on the stilt posts.

WARNING: Do not attempt to go under the machine unless the machine is secured by the overhead lifting means and the stilt posts have been welded in position to brace the base from further lowering.

3. Suspend a plumb bob through the center of the turntable's center shaft opening and check the base's position with respect to the center point mark on the pit floor. Make a note of which direction the base needs to be adjusted in order for the plumb bob to align. After the man under the base is out of the pit, reposition the base following his directions. Recheck the base's position with the plumb bob and adjust the base as necessary. As a double check, measure from the rim of the turntable to the pit wall to make sure that the turntable is centered within the pit.
4. Check the base's radial orientation in the pit by observing the pit-entrance platform section. This platform section should center-up and be square within the floor opening. Adjust base accordingly. Make sure that the turntable center remains aligned to the pit floor center point.

IMPORTANT: It is important that the two base strip anchor located under the offbear station align accurately with the 1" plates (E) on top of the stilt legs (B). If the strip anchors do not align with the plates, move the machine base until they locate properly upon the plates.

5. Check machine base for level.
6. After the machine base is level, if a gap exists between the base's bearing pads and any stilt post top, shim the gap as instructed on the foundation plan. It is important that the shimming procedure be followed carefully, especially at the strip anchors where a great upward force will be present. Do not final weld the base to the stilt post at this time, just tack-weld, in case it may be necessary to shift the machine later.
7. Adjust the 6" channel/6" post assembly under the base in the area of the pressurehead carrousel post. This assembly is designed for vertical height adjustment by sliding the 6" post legs inside the lower collars. Raise the assembly so the 6" channels bear solidly against the 3/8 plate on the machine's base. Make sure that the 6" channels do not obstruct the hole in the 3/8 base plate. The 6" posts can be welded fully as specified to the height adjustment collars, however, only tack weld the collars to the pit floor pads and the 6" channels to the 3/8 base plate.

8. Place the 8" perimeter channels into the pit and lay along side of the pit wall. Do not weld the channels to the wall. This will be done later. It is important to put these channels into the foundation now because it is not possible to place them into the pit once the machine's platform sections are installed.
9. After the base has been properly leveled, check if the turntable rim has remained centered within the pit wall. To prevent side shift of the machine, temporarily tack angles from the base's 8" channel structure to the I-beams imbedded into the pit wall.

INSTALL PLATFORM

1. Install the 10" channel platform support arms to the 8" platform post on the turntable. The splice plate on each support arm is match-marked to the corresponding splice plate on the 8" post. It is important that the correct support arm be fitted to the correct post. Use 3/4" diameter x 2" long bolts, lock washers, and nuts at the splices.
2. Install the (3) triangular platform segments. The splices on the center support pipe are match-marked to the corresponding platform segment splices in order for the segments to be installed at the correct station. It will be necessary to rotate the turntable so each platform segment can be installed at the machine's offbearing station. It may be necessary to go into the pit and loosen the turntable drive wheels if there is too much drag on them in order to rotate the turntable by hand. Use 3/4" diameter x 1-3/4" bolts and lock washers at the splice on the center support pipe. Use 7/8" diameter x 5-1/2" long bolts, flat washer, and locking nut to fasten the platform segments to the 10" channel platform support arms and 8" platform support pipes; with locking nut to the top. See detail A on installation illustration drawing for details. Tighten the locking nuts until the flat washers are snug. Each platform segment is bolted to the 10" channel and 8" pipe assembly at (4) points.

NOTE: Each platform segment has (2) 1-1/4" thick guide blocks that "guide" the platform segments to the 10" channel support arms. Do not remove these guide blocks from the platform segments because it will be very difficult to install the segments without these blocks. The blocks are to be removed in a later step.

3. Use 3/4" diameter x 1-3/4" bolts, lock washers and nuts between the 3/8" x 6" side plates of the platform segments.
4. Remove the (2) 1-1/4" installation blocks from under each platform segment.
5. Rotate the turntable, if not already done, so the turntable is locked into its exact station position. Advance the turntable lock bar so the lock bar inserts enter the slot at the rim of the turntable.
6. Rotate the carrousel into station position and advance the lock bar on the carrousel into the slot at the center pipe. Suspend a plumb bob from the carrousel jig strap at station center and check the plumb bob's position to the 1/4" hole at the turntable

station center. If required, adjust the 12" carrousel post so the plumb bob sets over the 1/4" station hole.

7. With the plumb bob in proper alignment at the turntable station center, tack weld the floor tie plate at the 12" carrousel post. See detail C on installation illustration drawing.
8. Unlatch the carrousel from the carrousel center pipe and rotate the carrousel and check the two remaining stations to the turntable station center. Make sure to lock the carrousel into position when checking with the plumb bob.

INSTALL CONCRETE FEEDER

1. Lift the feeder frame as specified by the handling drawing and position near its final location.
2. Lift the conveyor/hopper assembly and place atop the feeder frame. Bolt the assembly to the frame at the splice plates with 3/4" diameter bolts x 2" long lock washers and nuts. The hopper legs bolt to the frame top with 3/4" diameter bolts x 2" long lock washers and nuts.
3. Shim under the feeder frame so the conveyor and frame are level and so the conveyor's front splice plate is at the same elevation as the corresponding center pipe splice. Then slide frame up so splice plates align. Use 3/4" diameter x 3" long bolts, lock washers and nuts at the center pipe splice.
4. Bolt the fill station jig strap to the frame of the conveyor. The strap is match-marked so it can be correctly installed end-for-end.
5. From the jig strap, suspend a plumb bob down to the turntable. Check plumb bob to the 1/4" hole at the turntable station center. Adjust the feeder frame so the plumb bob aligns to the hole. Tack weld the frame legs to the weld pads in the floor. The legs will not be final-welded to the weld pads until checked by the Hawkeye technician.

NOTE: As a double-check to make sure that the feeder sets "square" to the pipe machine, measure from each front feeder leg to the center of the pipe machine's outer posts. These dimensions should be equal. See detail on machine component placement drawing.

INSTALL HEADER TRANSFER RAIL

1. It is important that the header cart transfer rail set "square" to the pipe machine. Refer to the machine component placement drawing for details on establishing a line for locating the transfer rail.
2. Position the header transfer rail near its final position as shown by machine component placement drawing. Advance the header cart forward and park under the carrousel. The cart should stop and bear against the rubber bumper at the front

of the rail. Adjust the bumper at its mid-position, so it can be adjusted equally in either a forward or reverse direction.

3. The header cart is strapped down tight to the wheel frame for shipment. It is necessary to unstrap the cart top so it is free to assume its normal operating position before locating the rail. Therefore, cut all shipping straps free of the wheel frame.
4. Suspend a plumb bob through the carousel station center jig strap that is located over the header cart top. Adjust the cart track so the plumb bob centers up over the 1/4" locating hole at the cart center. The transfer rail must be "square" to the machine. Make sure it sets parallel to the line marked in step #1 above.
5. Bolt the track to the floor and the track floor plates. Use concrete anchor bolts for this application.

FINAL ASSEMBLY

1. It is important that the machine base be tied into the pit wall to prevent torque or twisting of the machine upon the stilt posts. The stilt posts alone are not designed to resist side forces. Refer to the foundation drawing that shows the nominal elevation of the 8" channel structure that is to be placed around the inside perimeter of the pit wall. You should check this elevation with respect to your actual base elevation, and adjust the 8" perimeter channel length accordingly. Weld the 8" channel structure to the weld pads in the pit wall as specified by the foundation drawing.
2. Cut and place steel decking plates from the perimeter channel structure to the machine base. In several spots, tack weld the decking to the machine's 8" channel's base frame to the decking support channel. This is for safety reasons to prevent side shift of the machine. Final welding of the decking plates is to be done after the Hawkeye installation man has arrived.
3. Install the dip tank at the dip-tank station. Suspend a plumb bob from the carousel jig strap that is positioned over the dip tank station and mark the point on the floor where the plumb bob is located. Refer to machine component placement drawing to locate the dip tank from this point on the floor.
4. Install the safety railings around the carousel and header cart rail as specified by machine component placement drawing. Bolt these railings to the floor with concrete anchor bolts.
5. If purchased, install the ergonomic tables as per "Ergonomic Table Installation" drawing.
6. Install the presshead assemblies to each station on the carousel. Each assembly is connected to the carousel with three turn-buckles. Refer to detail D on machine installation illustration drawing for correct orientation of these assemblies to the carousel.
7. Install the railing at the pit entry floor plate. Bolt the railing to the platform structure with 3/4" diameter nuts.

8. Install the guard to the 12" carrousel support post. The guard bolts to the post with 3/8" diameter bolts x 1" long.

FINAL ASSEMBLY (Under Supervision of Hawkeye Technician Only)

1. Install hydraulic collector at center of turntable.
2. Install (2) center support pipes between base and floor.
3. Final weld points
 - a) stilt posts to base
 - b) pressurehead floor tie plate
 - c) machine base-to-pit wall decking
 - d) pit entry platform tie plates
 - e) platform safety wedges
 - f) feeder legs to weld pads
 - g) feeder table jib splice
4. Install feeder chute and turnhead ring.
5. Install rear chute to feeder.

EXTRUDER BEAM

1. The extruder beam has centering nibs welded near the center lift round. The purpose of these nibs is to keep your crane load hook centered on the lift round. Trim the nibs evenly so when your crane hook is centered over the lift round, a 1/8" gap is obtained between the hook and centering angles.
2. In some instances, the crane hook may be too large to fit inside the extruder beam. If this is the case, you must purchase an adaptor hook of the same capacity as the load hook that will fit the extruder beam. Connect the adaptor hook to the existing crane load hook with a shackle or similar method.
3. It is important that the overhead cranes upper limit switch be adjusted so the strip/extruder beams stop short of striking the crane bridge girders. If the strip/extruder beam would strike the bridge, it can cause damage or failure of the beam or crane, resulting in possible serious bodily injury to the crane operator and others.

To test to make sure that the upper limit switch is set properly, attach the strip/extruder beam to the crane hook and elevate the beam up near the crane bridge. Continue to slowly raise the strip/extruder beam to check to see if the upper limit switch stops the hoist's lifting motion before the strip/extruder beam can strike

the crane bridge. If the beam hits the crane bridge, re-adjust the limit switch so it does not.

SERVICE RUNS

The instructions that follow are for routing service runs to the pipe machine. These runs are from the customer's power sources to the machine. Some internal runs are required on the pipe machine. The internal runs are to be made under supervision of the Hawkeye technician. Runs from the customer's power sources are to be made by the customer.

NOTE: All machine connectors are JIC style. The customer is responsible for supplying adapter fittings for making connection to the JIC fittings.

HYDRAULIC

1. Set the hydraulic power unit in its proper setting. It is recommended that the hydraulic unit be placed in an isolated room, preferably lined with a noise dampening material to help reduce the noise level in the general working area. This room must be heated and well ventilated. Keep the unit as close as possible to the pipe machine in order to keep line runs and pressure losses to a minimum.
2. Run your hydraulic supply lines from the hydraulic unit to the machine fittings located inside the pit near the pit entry access. Your hydraulic lines should be connected to these fittings. See the service requirement drawings included with our installation packet for proper pipe size, pressure rating, etc.
3. Fill the hydraulic unit's reservoir with hydraulic oil. It is recommended that Rando oil HD68, or equivalent be used in the hydraulic system. See the attached specification sheet for details. Fill the reservoir so the fill gauge on the reservoir shows being approximately 2/3 full. Refer to the operating manual for procedures on adding oil.

IMPORTANT: Do not start the hydraulic unit. The hydraulic unit must be properly primed before starting or serious damage may occur to the unit. Also, supply lines must be flushed before they are connected to the pipe machine. The Hawkeye installation technician will prime and start the unit and assist in flushing out your lines. After your hydraulic lines are filled, it will probably be necessary to add additional oil to the hydraulic unit reservoir.

Electrical

1. Run the electrical power lines to the magnetic starter housing box on the hydraulic

unit. For power requirements, see the electrical service requirement drawing included in your installation packet.

2. Run the electrical power lines to the PipePro Machine. For power requirement at the machine, see the electrical service requirement drawing included with the installation packet.
3. Ground the frames for both the pipe machine and the hydraulic power unit. See the grounding instructions included with the installation packet.
4. An electrical control circuit is required between the PipePro Machine and the hydraulic power unit. Run electrical wire lines and conduit from the hydraulic power unit to the pipe machine as specified on the service requirement drawing.

NOTE: There is some internal wiring that must be completed before the PipePro Machine can be put into operation. These wire runs will be completed under supervision of the Hawkeye technician.

Air

1. Air can be routed to the PipePro Machine. Blow out all new lines leading to the machine before making final connection to the machine. An air shut off valve should be located on or near the machine for maintenance. For air requirements, see the service requirement drawing included with your installation packet. The air connection is near the pit entry access, same as the hydraulic connection.

Water

1. Route a water line to the PipePro Machine. For water requirements, see the service requirement drawing included with your installation packet. The water connector is in the pit entry access, near the air and hydraulic connections.